Surface Deformation Associated with the 2007 M_w8.1 Solomon Islands Earthquake ~~Cycle~~: A Joint InSAR and Coral Geodesy Analysis

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Key Points:

- Stacking solutions derived from interferogram subsets are used to distinguish deformation signals from tropospheric noise.
- ALOS InSAR observations capture meter-to-centimeter level deformation signals during and after the 2007 M_w8.1 Solomon Islands earthquake.
- Vertical deformation time series inferred from field coral δ^{13}C measurements can greatly extend surface deformation records back in time.

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Abstract

Spaceborne Interferometric Synthetic Aperture Radar (InSAR) observations and δ^{13}C measurements from coral microatolls are utilized to investigate surface deformation associated with the 2007 M_w8.1 Solomon Islands earthquake cycle. Due to limited field validation data, a new InSAR analysis method is designed to distinguish deformation signals from tropospheric noise based on stacking solutions derived from subsets of interferograms. InSAR results reveal up to -235.8 cm of line-of-sight (LOS) deformation during the earthquake, as well as a LOS deformation rate decay from 26.4 to 5.9 cm/year within $\sim$ 2 years after the earthquake. To further extend surface deformation records back in time, coral samples were collected from a *Porites lutea* microatoll grown in the western lagoon of Vonavona (locally known as Parara) Island. Vertical tectonic deformation time series (1996-2012) derived from coal Skeletal δ^{13}C measurements show a $\sim$ 2.3 cm/year subsidence trend over the ten year period prior to the earthquake, a 67 cm uplift during the 2007 earthquake, and $\sim$ 50 cm subsidence within the first 2.5 years after the earthquake. The characteristics of the temporal evolution of postseismic deformation inferred from independent coral and InSAR data are consistent. This is the first joint InSAR and coral analysis that spans ~~all phases~~ of an earthquake cycle, and their complementary use expands the temporal and spatial coverage of surface deformation observations over this active subduction zone.

Plain Language Summary

The Solomon Islands lie in one of the world's most active subduction zones, but its remote and difficult-to-access nature has limited the collection of high-quality surface deformation observations for studying tectonic activity. Utilizing a complementary analysis of imaging radar techniques and in situ coral samples, this study aims to quantify surface deformation prior to, during, and after the April 1, 2007 M_w8.1 Solomon Islands earthquake. A new radar imaging analysis algorithm is designed to distinguish deformation signals from tropospheric noise. Radar observations indicate up to -235.8 cm of deformation occurred during the earthquake along the radar LOS direction. Following the earthquake, LOS deformation rates decay from 26.4 to 5.9 cm/year within $\sim$ 2 years. Using the coral sample, nearly two decades of vertical deformation history is reconstructed. This time series reveals an $\sim$ 2.3 cm/year subsidence trend prior to the earthquake, 67 cm of uplift during, and $\sim$ 50 cm subsidence in the 2.5 years following the earthquake. The independent coral and radar data suggest consistent temporal evolution of the surface deformation after the earthquake. These data provide valuable information about the region's tectonic activity and fill the existing data gaps over this active subduction zone.

1 Introduction

Understanding the earthquake cycle is crucial for investigating the potential for future hazards (Scholz, 2002; Savage & Thatcher, 1992). The ~~interseismic~~ period is characterized by ~~the accumulation of strain~~ due to the locking of the faults, which ~~accounts~~ for ~~the longest portion of the cycle and often lasts~~ tens ~~to hundreds or thousands of~~ years. The coseismic period occurs as the accumulated ~~strain~~ exceeds the strength of the fault, and energy is released in the form of seismic waves in a few seconds or minutes. The postseismic period is defined by continued deformation stemming from both afterslip in the near field and various forms of relaxation in the far field over a few days to several months or years. Because the shallow megathrust (e.g., $<$ 20 km depth) is typically located beneath the ocean, it is not feasible to use land-based observations to study crustal deformation processes in such cases. However, the Solomon Islands, which was the site of a M_w8.1 earthquake that occurred on April 1, 2007, is an exception to this. Here, islands are located within several kilometers of the trench, offering a rare opportunity to quantify crustal deformation processes on the shallow megathrust. Despite this unique geological setting, the Solomon Islands are a data-scarce environment characterized by large gaps in observational data, a lack of in situ measurements, and a remote and difficult-to-access site. There are no permanent GPS stations on the New Georgia island chain near the epicenter and only two in the entirety of the Solomon Islands. Limited campaign GPS data exist; however, the duration of coverage is much shorter and the network is sparser than typical permanent networks.

Satellite remote sensing offers an opportunity to fill these observational gaps. Spaceborne Interferometric Synthetic Aperture Radar (InSAR) has been used to reconstruct millimeter to centimeter—surface deformations with tens to hundreds of meters of spatial resolution over broad areas (Massonnet et al., 1993). While InSAR has been successfully applied numerous times to investigate the earthquake cycle around the world (Wright et al., 2001; R. J. Walters et al., 2014; Delouis et al., 2002; Cakir et al., 2014; Kaneko et al., 2013; E. Lindsey et al., 2013; Lyons & Sandwell, 2003; Tong et al., 2013; Fialko et al., 2001; Johanson et al., 2005; Wei et al., 2015; Guns et al., 2022; Liu et al., 2018; Elliott et al., 2008; Bell et al., 2011; Wang & Wright, 2012; Wright et al., 2004; Ryder et al., 2007; Sijia et al., 2020; R. Walters et al., 2009; E. O. Lindsey et al., 2015), InSAR studies over the Solomon Islands are scarce. Dense vegetation leads to severe phase decorrelation noise even in C-band Sentinel-1 interferograms with short temporal baselines (e.g., 12 days). While L-band InSAR data are less prone to decorrelation, ALOS PALSAR data are the only L-band InSAR data with open access. Because the number of ALOS PALSAR acquisitions are limited and irregular ($\sim$ 10-15 SAR acquisitions over the entire mission duration of 2006-2010), it is very difficult to mitigate tropospheric turbulence noise (typically on the order of a few centimeters) due to the region's humid tropical climate. Furthermore, the availability of in situ measurements is extremely limited, making the validation of noisy InSAR results difficult. Miyagi et al. (2009) reported over 2 meters of LOS coseismic deformation from the 2007 M_w8.1 Solomon Islands earthquake based on ALOS PALSAR observations. Due to the difficulty in unwrapping interferograms over island regions, each of the three ALOS radar paths was processed individually. Miyagi et al. (2009) further solved for the slip distribution of this earthquake based on InSAR coseismic deformation estimates and limited field data, which suggested that the largest slip was around the hypocenter and centroid. Based on their slip model, they concluded that (1) the largest slip may be related to the subduction zone between the Woodlark and Australian plates, and (2) the smaller magnitude slip between the hypocenter and the centroid may be attributed to thermal activity from the volcano located on Simbo Island. To our knowledge, this is the only published InSAR surface deformation study related to the 2007 Solomon Island earthquake.

In this study, we show that L-band InSAR data can maintain phase coherence for $\sim$ 2 years over the densely vegetated Solomon Islands. This makes it possible to detect both coseismic and postseismic deformation caused by the 2007 Solomon Islands earthquake. Because postseismic deformation is much smaller than coseismic deformation, a major difficulty is the identification of InSAR phase signatures due to tropospheric noise that is comparable or larger than the signal of interest. Tropospheric noise consists of two components: (1) a stratified, topography-correlated component from variations in conditions in the vertical atmospheric column and (2) a turbulent and random component related to local weather fluctuations (Doin et al., 2009; Li et al., 2022). Although weather models and GNSS tropospheric zenith delay estimates have proven success in mitigating the stratified component (Doin et al., 2009; Yu et al., 2020), these external tropospheric correction layers often lack the spatial and temporal resolution to capture the localized turbulent component. To overcome this challenge, a new InSAR analysis method is developed to identify InSAR deformation signals from tropospheric noise based on their distinct temporal and spatial characteristics revealed by subsets of interferogram stacks. The method works well even with a limited number of high-quality SAR scenes. To further fill the existing observation gaps, this study also aims to advance the methodology for paleo-geodesy through the analysis of δ^{13}C measurements of coral skeleton samples in a given Porites microatoll located near the epicenter of the 2007 Solomon Islands earthquake. The vertical deformation time series inferred from both coral morphology and δ^{13}C geochemical analysis expands the temporal horizon of existing surface deformation studies back to 1996. This temporal expansion covers the interseismic period observational gap that exists with the ALOS data. This study is the first in which InSAR and coral measurements are utilized as complementary data sets.

2 Background

The Solomon Islands are situated astride an active plate boundary zone between the Australian and Pacific plates (Figure 1). Based on GPS campaigns, significant rates of oblique convergence ($\sim$ 10-11 cm/year) occur between these two plates (Demets et al., 1990; Wallace et al., 2004; M.-C. Chen et al., 2011). Additionally, the Woodlark microplate subducts beneath the north Solomons and New Britain. This microplate rotates clockwise, pulling away from the Australian plate, and thus further modifies the kinematics and subduction rates at the New Britain and San Cristobal Trenches (2.4°/Ma for 0.5 Ma to present) (Weissel et al., 1982; Benes et al., 1994; Wallace et al., 2004, 2014). The main boundary between the Pacific and Woodlark plates is the Woodlark basin, which is an active extensional basin where a new oceanic crust formation and seafloor spreading occurs, at rates of 4 to 5.6 cm/year in the eastern Woodlark basin (Wallace et al., 2014). Subduction of the Australian and Woodlark Plates occurs at the San Cristobal trench just west of the Solomon Islands at rates of $\sim$ 9-10 cm/year (Wallace et al., 2014) (Figure 1). The western Solomons, including the New Georgia island chain, are in close proximity to the San Cristobal trench. Ranongga, the island west of Kolombangara, is a mere 7 kilometers from the trench (T. Chen et al., 2009), and the western coast of Rendova is only $\sim$ 20 kilometers from the trench. Interestingly, Simbo Island, just northwest of the island chain of interest, lies directly atop the subducting plate. These characteristics further solidify the Solomon Islands as an ideal laboratory for studying surface deformation from spaceborne InSAR data.

Furthermore, the presence of coral microatolls fringing the islands provides an alternative way to reconstruct vertical deformation history in this tropical region (Taylor et al., 1987). Coral species grow both upward and horizontally; however, upward growth is limited by water depth (exposure to air and direct sunlight). This limit is known as the highest level of survival (HLS). During sustained emergence above the ocean surface, exposed coral surfaces die and produce notable die-downs in the coral morphology, and growth continues laterally and upward at the HLS. When relative sea level rises (or the ocean floor subsides), coral colonies grow vertically toward the sea surface at rates ranging from 1 to 1.5 cm/yr. A striking example of such growth and die-downs is provided by the M_w8.1 earthquake of April 1, 2007, which produced dramatic vertical coastal displacements. Taylor et al. (2008) utilized coral microatolls to investigate the catastrophic event that ruptured just west of the New Georgia island chain. They analyzed the vertical coral growth and die-downs to measure the associated coastal coseismic subsidence and uplift, enabling some of the few coseismic deformation measurements within several kilometers of the trench. Their field surveys documented 0.73 $\pm$ 0.14 m of coseismic subsidence at Simbo Island, and up to 2.46 $\pm$ 0.14 m of coseismic uplift at Ranongga Island, located only 8 km away on the overriding plate. Their results revealed that the 2007 earthquake ruptured across a triple junction where the Australian and Woodlark plates subduct beneath the overriding Pacific plate. Similarly, coral microatolls have been used to study coseismic vertical deformation in other regions that have experienced large magnitude earthquakes, including Haiti and Sumatra (Zachariasen et al., 1999; Briggs et al., 2006; Meltzner et al., 2010; Weil-Accardo et al., 2016).

3 Method

3.1 InSAR Data Processing

ALOS PALSAR scenes acquired between December 28, 2006 and January 22, 2010 over the area of interest (Figure 2a) were downloaded from the Alaska Satellite Facility (Table S1). All SAR scenes were processed using a motion-compensation imaging radar processor (H. A. Zebker et al., 2010), and no spatial or temporal baseline constraints were applied during interferogram formation. All interferograms were multi-looked to 240-meter pixel spacing, and the topographic phase component was removed using a Digital Elevation Model (DEM; 30 m pixel-spacing) derived from the Shuttle Radar Topography Mission (SRTM) (Farr et al., 2007). Interferograms acquired on the same day from the same path were merged by minimizing the phase difference between these overlapping regions of the adjacent frames, and the Goldstein filter was applied to merged interferograms to improve spatial phase coherence (Goldstein & Werner, 1998).

These interferograms were then unwrapped using Statistical-cost, Network-flow Algorithm for Phase Unwrapping (SNAPHU) algorithm (C. W. Chen & Zebker, 2001). SNAPHU attempts to unwrap InSAR phase measurements at all pixels including bodies of water where phase is completely decorrelated and random in space. This can dramatically increase computing time and introduce unwrapping errors at nearby land pixels. To overcome this challenge, land and ocean pixels were classified using an average phase coherence threshold of 0.11, and then wrapped phase values at ocean pixels were replaced by wrapped phase values at nearby land-pixels (e.g., Shanker and Zebker (2009)). As shown in Figure S1, the wrapped phase value ϕ_{ocean} at an ocean pixel of interest was replaced by the weighted sum of wrapped phase values at nearby land pixels as:

$$\phi_{ocean} = \frac{\sum w_i \phi_i}{\sum w_i} \tag{1}$$

where ϕ_i is the wrapped phase of the i^{th} land pixel within a square box of length 11 pixels centered at the ocean pixel, $w_i = 1/r_i$, and r_i is the distance between the ocean pixel of interest and the i^{th} land pixel within the square box. This phase filling process was performed for a second time to further extrapolate phase observations at land pixels into ocean area, which reduces large phase discontinuities at the coastline and phase unwrapping errors at land pixels. After unwrapping, the land-ocean mask was employed to identify land pixels for surface deformation analysis.

At a land pixel of interest, InSAR unwrapped phase ϕ measures surface deformation Δd_{LOS} along the radar line-of-sight direction with respect to a reference pixel as:

$$\phi \approx \frac{4\pi}{\lambda}\Delta d_{LOS} = \frac{4\pi}{\lambda}(e_1\Delta d_e + e_2\Delta d_n + e_3\Delta d_u) \tag{2}$$

where λ is the radar wavelength. Δd_e, Δd_n, and Δd_u are surface deformation in east, north and vertical directions, and $\mathbf{e} = [e_1, e_2, e_3]$ is the LOS unit vector. For the ALOS PALSAR data over the Solomon Islands, $\mathbf{e} = [0.6, 0.13, -0.79]$ at the mid-swath, and the look angle variation is not substantial across the ALOS PALSAR scenes. Coseismic deformation measurements (in east, north, and vertical directions) of the April 1, 2007 Solomon Islands earthquake were recorded at three campaign GPS stations RAVA, MUND, and HUSU (Figure 2a). The GPS station RAVA (8.721°S 157.407°E) was chosen as the reference point to calibrate the InSAR coseismic deformation map. The other two stations were used to validate InSAR results. Postseismic GPS surface deformation time series were recorded at MUND for a brief period from April 23, 2007 through May 5, 2007. However, the limited data acquisition period makes it impossible to capture the post-seismic deformation rate change over the ALOS PALSAR study period. Therefore, post-seismic interferograms were not calibrated to campaign GPS data. Instead, InSAR post-seismic deformation rate estimates were calculated with respect to a local reference point (8.202°S 157.355°E) on the New Georgia island away from the epicenter and the coast.

3.2 Coseismic and Postseismic LOS deformation solutions

As shown in Equation (2), InSAR calculates the phase difference between two SAR acquisitions, from which the LOS surface deformation between these two acquisition dates can be derived (Rosen et al., 2000). Given that earthquakes occur over a short period of seconds to minutes and the ALOS satellite revisit time is 46 days (the acquisitions were irregular with data gaps), a coseismic deformation signal can be modeled as a step function (Figure 3a), and can be captured by interferograms that span the earthquake. Here one coseismic interferogram was formed using two Path 344 SAR scenes acquired on March 1, 2007 and April 16, 2007. Another coseismic interferogram was formed using two Path 343 SAR scenes acquired on February 12, 2007 and May 15, 2007. The two-frame interferograms were then merged to generate a single 4-frame coseismic interferogram. The magnitude of the expected coseismic deformation signal in the near-field of the event is on the order of meters (Miyagi et al., 2009; Taylor et al., 2008), which is well above the InSAR noise level. As a result, averaging multiple interferograms that span the April 1, 2007 Solomon Islands earthquake is not necessary.

The postseismic signal is expected to follow an exponential or logarithmic decay (Figure 3b) depending on the postseismic mechanism at play (Sobrero et al., 2020; Yusiyanti et al., 2023). Because the magnitude of the postseismic deformation is significantly smaller than the coseismic deformation, a stacking approach (Sandwell & Price, 1998; Staniewicz et al., 2020) is employed to calculate the average LOS velocity, $\bar{v}_k$, for the k^{th} period of interest after the 2007 Solomon Islands earthquake:

$$S_k = \frac{\lambda}{4\pi}\frac{\sum_{i\epsilon G_k}\phi_i}{\sum_{i\epsilon G_k}\Delta t_i} = \frac{\sum_{i\epsilon G_k}\Delta d_{LOS,i}}{\sum_{i\epsilon G_k}\Delta t_i} + \delta = \bar{v}_k + \delta \tag{3}$$

where S_k denotes the k^{th} stack, $\lambda = 24$ cm is the ALOS PALSAR radar wavelength, and G_k is a set of interferograms that were formed using two SAR scenes acquired within the k^{th} period of interest. ϕ_i, $\Delta d_{LOS,i}$, and Δt_i are the unwrapped phase, the corresponding LOS deformation, and the temporal baseline of the i^{th} interferogram in G_k, respectively. δ is the residual noise term.

Nine ALOS PALSAR path 344 scenes acquired between April 16, 2007 and January 19, 2010 were used in the postseismic deformation analysis (Figure 3b). Here the postseismic deformation analysis is limited to every land pixel in path 344. This is because path 344 is closer to the 2007 earthquake epicenter than path 343, and the expected signal-to-noise ratio (SNR) is better. Following Equation (3), the average LOS velocity was calculated over three consecutive periods of interest: (1) April 16, 2007 through October 17, 2007; (2) October 17, 2007 through April 18, 2008; and (3) April 18, 2008 through January 22, 2010. The interferograms used in each consecutive stack are listed in Table 1. Furthermore, the average LOS velocity was calculated over the following growing time windows starting on April 16, 2007 through: (1) September 1, 2007 (2 acquisitions and 1 interferogram); (2) October 17, 2007 (3 acquisitions and 3 interferograms); (3) January 17, 2008 (4 acquisitions and 6 interferograms); (4) March 3, 2008 (5 acquisitions and 10 interferograms); (5) April 18, 2008 (6 acquisitions and 15 interferograms); (6) September 3, 2008 (7 acquisitions and 21 interferograms); (7) January 19, 2009 (8 acquisitions and 28 interferograms); and (8) January 22, 2010 (9 acquisitions and 36 interferograms). Section 3.3 further presents a method to determine whether postseismic deformation signals derived from these stacks of interferograms are above the residual noise level and thus detectable.

3.3 InSAR Measurement Uncertainty Quantification

The unwrapped InSAR phase at a pixel of interest, ϕ , can be written as:

$$\phi = \frac{4\pi}{\lambda}\Delta d_{LOS} + \Delta\phi_{DEM} + \Delta\phi_{decorr} + \Delta\phi_{unwrap} + \Delta\phi_{orb} + \Delta\phi_{iono} + \Delta\phi_{tropo} + \Delta\phi_n \tag{4}$$

where λ is the radar wavelength, and Δd_{LOS} is the LOS deformation between the two SAR acquisitions used to form the interferogram. The remaining terms are the result of various error sources: inaccurate DEM ($\Delta\phi_{dem}$), phase decorrelation ($\Delta\phi_{decorr}$) and the resulting unwrapping errors ($\Delta\phi_{unwrap}$), orbital errors ($\Delta\phi_{orb}$), ionospheric ($\Delta\phi_{iono}$) and tropospheric ($\Delta\phi_{tropo}$) artifacts, and other smaller error terms ($\Delta\phi_n$) such as changes in soil moisture.

Given that the magnitude of the observed InSAR phase does not correlate with InSAR perpendicular baseline, a key signature of DEM errors (Fattahi & Amelung, 2013), the impact of DEM errors is minimal. Based on visual inspection, all interferograms formed using path 344 scenes acquired on two dates (January 14, 2007 and October 19, 2008) suffer from severe phase decorrelation artifacts and relatively large unwrapping errors. This is likely due to changes in surface scattering properties on these two days (e.g., due to a severe weather event). These two SAR acquisitions and the associated interferograms were excluded, and a total of nine high-quality path 344 SAR acquisitions were used in the postseismic deformation analysis (Figure 3b). For all interferograms formed using these nine path 344 acquisitions, InSAR phase coherence is well maintained regardless of temporal baselines, except over Tetepare Island, which lies southeast of Rendova. Consequently, the postseismic deformation analysis focuses on Vonavona (locally known as Parara), Kohinggo, Kolombangara, and New Georgia, where decorrelation artifacts $\Delta\phi_{decorr}$ and unwrapping errors ($\Delta\phi_{unwrap}$) are not substantial. A planar phase ramp was estimated and removed in each postseismic interferogram to account for orbital errors ($\Delta\phi_{orb}$) and long-wavelength tropospheric and ionospheric noise. Because the orientation of the observed phase ramp in each interferogram varies, these ramps are unlikely related to true deformation signals. The other error sources, $\Delta\phi_n$, associated with thermal and soil moisture effects, are on the order of millimeters and much smaller than other noise terms (Michaelides et al., 2019). Thus, it follows that the dominant error term is tropospheric noise in the Solomon Islands ALOS PALSAR data set, and Equation (4) can be simplified to:

$$\phi_{ij} = \frac{4\pi}{\lambda}(\Delta d_{ij} + n_j - n_i) \tag{5}$$

where λ is the radar wavelength and Δd_{ij} is the LOS surface deformation between two SAR acquisitions acquired on dates i and j. The tropospheric noise in the LOS direction on each acquisition date is represented as n_i and n_j, respectively.

Based on nine ALOS PALSAR path 344 scenes (Figure 3b), consider the stacking results over three consecutive periods of interest: (1) April 16, 2007 through October 17, 2007; (2) October 17, 2007 through April 18, 2008; and (3) April 18, 2008 through January 22, 2010. Following Equation (3) and Equation (5), each of the three stacks defined in Table 1 can be expressed as:

$$S_1 = \bar{v}_1 + \frac{2n_3 - 2n_1}{\sum\limits_{i=1,i<j}^{3} \Delta t_{ij}} \tag{6}$$

$$S_2 = \bar{v}_2 + \frac{3n_6 + n_5 - n_4 - 3n_3}{\sum\limits_{i=3,i<j}^{6} \Delta t_{ij}} \tag{7}$$

$$S_3 = \bar{v}_3 + \frac{3n_9 + n_8 - n_7 - 3n_6}{\sum\limits_{i=6,i<j}^{9} \Delta t_{ij}} \tag{8}$$

Here $\bar{v}_1$, $\bar{v}_2$, and $\bar{v}_3$ are the average deformation rates for each of the three periods of interest. n_i is the LOS tropospheric noise of the i^{th} SAR acquisition ($i = 1, \cdots, 9$), and the total time span of all interferograms ($\sum \Delta t_{ij}$) in the three stacks are 1.01, 1.64, and 5.67 years, respectively. In the case where postseismic deformation signals are well above the noise level, a consistent deformation pattern should be observed in all three stacking solutions. At a given location, the observed deformation rate is expected to decrease in time (e.g. a larger rate is expected in S_1 than S_3). For a given stack solution S_k, the observed deformation rate is expected to decrease as the distance to the epicenter increases. In contrast, tropospheric errors are not correlated for time increments greater than one day (H. A. Zebker et al., 1997; Emardson et al., 2003). Given that the ALOS PALSAR satellite revisit cycle is 46 days, $n_1, \cdots, n_9$ are temporally uncorrelated. This suggests that the residual tropospheric noise patterns in S_1 and S_3 are not correlated. The residual noise level in S_3 tends to be smaller than in S_1 because the total time span of all interferograms in S_3 (5.67 years) is more than five times larger than in S_3 (1.01 years). Furthermore, the third SAR acquisition (October 17, 2017) was used to form interferograms in both S_1 and S_2. As shown in Equation (6) and Equation (8), the contribution of n_3 in S_1 and S_2 has the opposite sign. Similarly, the contribution of n_6 in S_2 and S_3 also has the opposite sign. In summary, the temporal and spatial characteristics of tropospheric noise and postseismic deformation signals are different. This allows determination of whether postseismic deformation signals are detectable by a comparison of S_1, S_2, and S_3.

Next, consider the following growing time windows starting on April 16, 2007 through: (1) September 1, 2007 (2 acquisitions); (2) October 17, 2007 (3 acquisitions); (3) January 17, 2008 (4 acquisitions); (4) March 3, 2008 (5 acquisitions); (5) April 18, 2008 (6 acquisitions); (6) September 3, 2008 (7 acquisitions); (7) January 19, 2009 (8 acquisitions); and (8) January 22, 2010 (9 acquisitions). For the case of M SAR acquisitions that all $N = M \times (M-1)/2$ interferograms are formed, the stacking result, S_M, can be written as:

$$S_M = \bar{v}_M + \frac{1}{\sum\limits_i^N \Delta t_i} \left(\sum_m^M w_m n_m\right) \tag{9}$$

where $\bar{v}_M$ is the average deformation rates for the M^{th} time window, and $\sum\limits_i^N \Delta t_i$ is the total time span of all interferograms in the subset. n_m is the tropospheric noise of the m^{th} SAR acquisition, and the weight applied to the tropospheric noise of the m^{th} SAR scene, w_m, can be described as

$$w_m = 2m - (M+1) \tag{10}$$

Equation (9) and Equation (10) quantitatively describe the stacking results of eight growing time windows, where each stack contains an additional SAR scene forward in time. For $M = 2$, $w_1 = -1$ and $w_2 = 1$. For $M = 9$, the largest absolute weights are $w_1 = -8$ and $w_9 = 8$. Furthermore, as M increases, the total time span of all interferograms increases. For the Solomon Islands case, the total time space for $M = 2$ is 0.378 years, while the total time space for $M = 9$ is 34.5 years. This means that the residual tropospheric noise from the 1^{st} SAR acquisition (April 16, 2007) is more than 11 times larger when $M = 2$ than when $M = 9$. In this growing time window stacking design, both the postseismic deformation signal and n_1 are expected to produce a spatially consistent pattern with reduced magnitude over time. In contrast, residual tropospheric noise from the remaining SAR acquisitions is expected to produce random patterns in these eight stacking results. To further separate the contribution from the postseismic deformation signal and n_1, an alternative stacking test excluding all interferograms formed using the 1^{st} SAR acquisition (April 16, 2007) can be calculated, where n_1 does not contribute to the result.

3.4 Time Series of Vertical Displacements Derived from Coral δ^{13}C Measurements

The ALOS PALSAR mission only captured two acquisitions prior to the 2007 earthquake, and one of these two was excluded in InSAR analysis due to severe phase decorrelation. Consequently, years of the interseismic period cannot be reconstructed from InSAR data alone. In order to fill the interseismic data gap, this study further investigates the δ^{13}C (^{13}C/^{12}C) signal preserved in shallow-living coral microatolls and its ability to expand the temporal horizon of surface deformation records to nearly two decades. Vertical deformation derived from δ^{13}C measurements tends to be more accurate than coral morphology observations. This is because microatolls do not always emerge above the water during sea-level drops, and growth may not keep pace with rapid sea-level rise (Philibosian, 2025).

Previous studies have established a linear relationship between coral skeletal δ^{13}C and water depth, with decreasing δ^{13}C values at greater depth. This is attributed to reduced light penetration, lower photosynthetic activity, and declining irradiance with depth (Weber & Woodhead, 1970; Weber et al., 1976; Swart, 1983). Based on this relationship, δ^{13}C acts as a proxy for relative sea level (RSL), which is influenced by both tectonic activity and climate patterns. Here climate signals often contain annual variations and some semblance of cycles (e.g., El Nino–Southern Oscillation), whereas tectonic signals behave differently – interseismic deformation often appears as a long-term trend in time series, and postseismic deformation often follows the expected decay model. Furthermore, the relative sea level history can be compared with independent measurements, such as tide gauge data, to further isolate vertical deformation due to tectonic activity (Gagan et al., 2008, 2009).

The coral cross-section used in this study (Figure S2) was retrieved in April 2012 from a *Porites lutea* microatoll that had grown in the western lagoon of Vonavona (locally known as Parara) Island in the Solomon Islands (8.221° S, 157.005° E). The coral slab measured 138×100×6 cm, and its morphology records multiple exposed surfaces associated with emergence events during relative sea-level drops. All exposed surface measurements for this study include both the length of the dead surface and the depth of the water above the coral top at the time of collection. X-ray scans of the coral slab (Figure 4) reveal clear annual growth rings, analogous to tree rings. By counting the growth rings backward into time starting from the date of collection, historic events can be dated with annual or better temporal resolution (Buddemeier & Taylor, 2000; Biemiller et al., 2020). For example, the April 2007 M_w8.1 megathrust earthquake caused an uplift that killed the living coral surface (marked by the green line in Figure 4) (Taylor et al., 2008). Following this 2007 exposed surface line, samples were collected using a computer-controlled drill along 7 transects (A–G) (marked by the yellow lines in Figure 4), each perpendicular to the growth rings. Each transect consisted of a series of about 10 single samples spanning a total of 50-60 mm in vertical direction ($\sim$ 3 years of coral growth). A total of 68 samples from Transects A-G, each with a drilled volume of 1.5×1.5×1.0 mm, were obtained, and δ^{13}C were analyzed and recorded to estimate the linear relationship between δ^{13}C and water depth.

To further reconstruct δ^{13}C time series over longer time scales, a horizontal transect (H) spanning the years 1996–2012 (marked by the red line in Figure 4) was sampled and analyzed. A total of 118 samples along transect H enables reconstruction of inter-, co-, and postseismic vertical motion of the microatoll using the linear δ^{13}C-water depth relationship derived from the vertical sampling analysis. To isolate skeletal δ^{13}C signals associated with climate and tectonics, GPS-corrected tide-gauge records from Honiara (9.429° S, 159.955° E; Fig. 2b) were employed (Caldwell et al., 2001). Sea surface height measured by the satellites using radar altimetry (Zlotnicki et al., 2019) reveals a strong correlation between the study site with the tide gauge in Honiara and Bougainville. This means that the absolute sea levels at Vonavona and Honiara are similar. However, Honiara is too far away from the epicenter, and deformation signals related to the 2007 earthquake cycle are not detectable here. The relative sea level history derived from the δ^{13}C measurements acts as a complementary data set that shows consistency with remote sensing observations. It is important to note that InSAR LOS measurements cannot be directly compared to vertical measurements from coral samples acquired at a different location. Nonetheless, consistency in the temporal evolution of the deformation and the order of magnitude can be analyzed as an alternative way of validation.

4 Results and Discussion

4.1 Coseismic Deformation Determined from InSAR

The 2007 Solomon Islands earthquake deformation map derived from the ALOS PALSAR coseismic interferograms is shown on a circular color scale in Figure 5. Here each fringe cycle (e.g., green to pink to yellow to green) indicates 20 cm of positive LOS deformation. In this study, the LOS direction is defined as from the satellite to the ground. Therefore, uplift or westward motion leads to negative LOS deformation in ALOS PALSAR ascending geometry, whereas subsidence or eastward motion leads to positive LOS deformation in the same geometry. Spatially continuous fringe patterns are observed across multiple islands. This suggests that the proposed phase-filling strategy can unwrap InSAR phase measurements over land surface pixels isolated by ocean pixels successfully. The observed fringes over Vonavona and Kohinggo are the densest in space, because these two islands are closest to the epicenter (Figure 2). The same coseismic LOS deformation map is also presented on a continuous color map (Figure 5b), which highlights a maximum magnitude of -235.8 centimeters of LOS deformation measured on the southwestern coast of the Vonavona island, approximately 26 kilometers from the epicenter. Kohinggo, another western island near the epicenter, experienced LOS deformation ranging from -100.1 to -194.4 cm. In contrast, Kolombangara recorded LOS deformation ranging from -48.2 to -128.6 cm, and Rendova recorded LOS deformation ranging from 11.8 to -69.5 cm. The spatial deformation gradient is visible on the large New Georgia island. The northwestern portion of the island near Kohinggo experienced up to -132.4 centimeters of negative LOS deformation during the earthquake, while the southeastern region experienced up to 18.6 centimeters positive LOS deformation. Vangunu, the island farthest east in the study region, measured up to 25.6 centimeters positive LOS deformation. This value is of significantly smaller magnitude than that of Vonavona, as expected because of the loss of energy in the seismic waves as they travel farther away from the epicenter.

Three GPS campaign stations are located within the coseismic map in Figure 5: MUND, RAVA, and HUSU. RAVA was used as the reference point to calibrate InSAR coseismic map, and the misfits between the InSAR LOS and the MUND and HUSU GPS stations are 3.2 and -3.9 centimeters, respectively (Table 2). The misfit values are within the expected tropospheric noise range of 2 to 5 centimeters of a single interferogram over humid tropical islands (J. Chen et al., 2014). Additionally, the coseismic interferograms generated from different paths acquired on different dates were merged without any visual discontinuities across path boundaries. This further confirms that the coseismic deformation signal is well above the noise level. Previous research reported up to 2.46 $\pm$ 0.14 m of coastal coseismic uplift at Ranongga island based on surveys of coral reef die-downs (Taylor et al., 2008). In comparison, the maximum LOS deformation at the nearby Vonavona island is 235.8 centimeters. The order of the deformation (meter-level) and the spatial patterns are consistent between spaceborne InSAR and field coral observations. We note that it is not possible to validate InSAR co-seismic deformation estimates directly using field observations. This is because (1) InSAR measures surface deformation along the LOS direction, while coral surveys measure vertical deformation. Here the decomposition of InSAR LOS deformation into horizontal and vertical components is not possible Due to the lack of multiple LOS imaging geometries; and (2) coral grows in the ocean, and InSAR can only measure surface deformation at land pixels. Thus there are no co-located InSAR and coral measurements.

4.2 Postseismic Deformation Determined from InSAR

Figure 6 shows the average LOS rate (in cm/year) over three consecutive time periods of interest: (a) the first six months following the earthquake from April through October 2007, (b) the following seven-month period covering October 2007 through April 2008, and (c) the remaining period for which SAR acquisitions are available that spans April 2008 through January 2010. In the green dashed box, the maximum LOS rate is observed near the epicenter on Vonavona with a similar spatial pattern for all three time periods. At a given location, the observed LOS rate decreases over time. This is consistent with the temporal characteristics of the postseismic deformation signals. In contrast, the LOS patterns over the island regions outside the green dashed box are correlated in space but random in time – characteristics that align with tropospheric turbulence noise behavior as discussed in Section 3.3.

Figure 7 shows the average LOS deformation rates of eight overlapping time windows starting on April 16, 2007 through: (1) September 1, 2007; (2) October 17, 2007; (3) January 17, 2008; (4) March 3, 2008; (5) April 18, 2008; (6) September 3, 2008; (7) January 19, 2009; and (8) January 22, 2010. The first stack (Figure 7(a)) includes only a single interferogram (2007/04/16-2007/09/01), which contains the largest postseismic deformation signal during the first 5 months after the earthquake. Inside the green dashed box (over Vonavona), the observed LOS rate decreases over time (Figures 7(b)-(h)), and the spatial patterns (larger deformation rates are at pixels closer to the epicenter) are consistent in all eight stacks. Based on Equation (9) and Equation (10), both the postseismic deformation signal and tropospheric noise on the first SAR acquisition (April 16, 2007) may produce the pattern inside the green dashed box in Figure 7. Nonetheless, Figure 6(b) - (c) also contain this spatially consistent LOS pattern inside the green dashed box, and these two stacks do not include interferograms formed using the first SAR acquisition. Therefore, the observed LOS pattern over Vonavona is likely associated with postseismic deformation signals rather than tropospheric noise. On the other hand, the observed LOS pattern over the other islands –Kolombangara, Kohinggo, and New Georgia– are correlated in space but random in time. This is consistent with the finding of Figure 6 that there are no detectable signals above tropospheric noise level here. As shown in Section 3.3, the residual tropospheric noise decreases as the total time span of an interferogram stack increases (Figures 7(a)-(h)).

Based on ALOS InSAR stacking solutions, the average LOS rate on Vonavona from April 16, 2007 through (1) September 1, 2007, (2) October 17, 2007, (3) January 17, 2008, (4) March 3, 2008, (5) April 18, 2008, (6) September 3, 2008 are 26.4 cm/year, 19.5 cm/year, 14.2 cm/year, 12.7 cm/year, 10.5 cm/year, and 6.2 cm /year, respectively. The average LOS rate over two longer periods, spanning April 16, 2007 through January 19, 2009 and January 22, 2010, are both at 5.9 cm/year. Our results follow the expected exponential decay behavior associated with postseismic afterslip (Perfettini & Avouac, 2004), which highlights the effectiveness of our algorithm for capturing the temporal evolution of surface deformation from sparse InSAR data set.

4.3 Vertical deformation from Coral δ^{13}C Time Series

The paleo geodesy coral analysis focuses on the 1997–2012 period, which includes three phases: (1) the interseismic period prior to the 2007 earthquake, (2) coseismic emergence during the earthquake, and (3) the postseismic period after April 1, 2007. Based on the morphology between the rings corresponding to years 1997 and 2007 in the coral skeleton (Figure 4), prior observations inferred that the microatoll subsided a minimum of ~ 17 cm in a 20-year period, was uplifted approximately ~ 63 cm during the earthquake, and subsequently subsided again by tens of centimeters. The coral scan in Figure 4 indicates a coseismic uplift extent of ~ 73 cm from the die-down surface associated with the 2007 earthquake (marked by the green line). This estimate is derived by combining the vertical extent of the 2007 die-down surface (63 cm) with the measured water depth (10 cm) above the coral top at the time of collection. The corresponding change in δ^{13}C is 2‰, yielding an estimated δ^{13}C-water depth relationship of ~ 36.5 cm/‰ from the morphology. To further establish the linear water depth relationship, samples from Transects A-G were used to quantify the relationship between water depth and δ^{13}C. The results (Figure 8) show a linear trend as reported by previous studies (Weber & Woodhead, 1970; Weber et al., 1976; Swart, 1983): skeletal δ^{13}C values decrease with increasing water depth. The root-mean-square error (RMSE) indicates additional variability, which is likely associated with sub-annual to annual sea-level changes. Nonetheless, the mean slope of 37.67 cm/‰ closely matches the coseismic estimate of 36.5 cm/‰ derived from the coral morphology, confirming that δ^{13}C is a reliable proxy for relative sea-level reconstruction.

The three phases of the earthquake cycle were further investigated through the relative sea level history reconstructed from the δ^{13}C measurements from Transect H and the corresponding water-depth relationship (Figure 8). The unfiltered relative sea level time series derived from the δ^{13}C measurements of coral samples is shown in Figure 9a. A clear vertical displacement, indicated by the sharp change in relative sea level, is present during the time of the 2007 Solomon Islands earthquake. This coseismic uplift indicated from the δ^{13}C measurements of the study sample (8.221°S, 157.005°E) is approximately ~~67 cm~~. A vertical measurement of the visible die-down in the morphology a nearby coral sampling site (8.227°S, 157.012°E) from a previous study (Taylor et al., 2008) recorded uplift of 50 ± 15 cm, highlighting the agreement between the two in situ measurement methods. An L1 linear regression is applied to the time series prior to the earthquake to determine the interseismic subsidence rate. The slope of the ~~interseismic~~ subsidence is approximately 2.3 centimeters per year. The Honiara tide gauge data (Figure 2b) are shown in Figure 9b. The two RSL data sets both capture similar seasonal weather signals and climate events (e.g. El Nino). However, the changes in relative sea level derived from δ^{13}C also contain contributions from tectonic motion associated with the 2007 Solomon Island earthquake. As such, the 2007 earthquake is not expected to influence the tide gauge measurements recorded at Honiara, because it is too far away from the epicenter. The interseismic subsidence trend in the δ^{13}C-derived relative sea level is not present (e.g., as a sea level increase) in the tide gauge data, further highlighting that this signal was the result of tectonic motion rather than regional sea level rise trends.

Following the coseismic uplift, there is subsidence that demonstrates exponential decay behavior during the post-seismic period. This post-seismic relative sea level is plotted with the InSAR LOS deformation over the eight time windows discussed above in Section 4.2 in Figure 10. The LOS deformation rate over each growing time window is known from the stacking framework. The total deformation was computed by integrating the deformation rates through time using a trapezoidal rule applied to the overlapping deformation rate estimates. The deformation for the initial interval (time between the first and second postseismic SAR scene acquisitions) was obtained by multiplying the corresponding deformation rate by the time duration between scenes. For each subsequent deformation rate, incremental deformation was calculated using the mean of consecutive deformation rate estimates multiplied by the additional time step associated with the newest SAR scene in the growing window and added cumulatively to the previous deformation. This approach yields a piecewise trapezoidal approximation of the total deformation through time. The similar temporal evolution between the LOS deformation and the coral-derived relative sea level highlights the consistency between the two independent data sets. This agreement among the δ^{13}C measurements and remote sensing observations supports the conclusions drawn from the InSAR-derived deformation.

5 Conclusion

In this study, surface deformation prior to, during, and after the 2007 Solomon Islands earthquake was reconstructed using InSAR and coral geodesy. Phase filling allowed multiple islands from two separate ALOS PALSAR paths to be successfully unwrapped as a single interferogram without discontinuities or visible unwrapping errors. The coseismic InSAR map was calibrated and validated using coseismic displacements from three campaign GPS stations, which is consistent with previously published results (Miyagi et al., 2009). A new InSAR analysis method based on stacking subset of interferograms was developed. This enabled, for the first time, the generation of a postseismic InSAR time series of the 2007 Solomon Islands earthquake. δ^{13}C measurements from a coral sample near Vonavona provided a record of the relative sea level in the region, which contained a signal from both oceanographic changes and vertical tectonic motion. Using a control data set from a nearby tide gauge, the contribution from the absolute sea level changes was quantified. Results suggested that the coral time series captured an interseismic subsidence trend, a 67 cm coseismic uplift, and a decay of subsidence rates following the earthquake. The characteristics of the temporal evolution of postseismic deformation inferred from independent coral and InSAR data are consistent. While our InSAR analysis method works with a limited number of SAR data sets–only 10 high-quality ALOS PALSAR scenes are available–this method can be used for robust uncertainty quantification, when a large number of SAR scenes become available (M. S. Zebker et al., 2023; M. S. Zebker & Chen, 2024). The study highlights the potential of L-band InSAR data for observing surface deformation over densely vegetated and humid regions. With the combination of an L-band sensor for penetrating dense vegetation and the shorter revisit time of 12 days, NISAR will produce high-quality InSAR observations over undersurveyed subduction zones. Furthermore, the results from the coral δ^{13}C measurements show promise as a method to greatly expand the time horizon of vertical deformation history back in time and complement modern geodesy observations.

Open Research Section

ALOS PALSAR SAR imagery over the New Georgia island chain in the Solomon Islands (Paths 343 and 344, Frames 7010 and 7020) can be downloaded from the Alaska Satellite Facility at https://search.asf.alaska.edu. All interferograms in this study were processed using a motion-compensation imaging radar processor (H. A. Zebker et al., 2010). Comparable interferograms can be generated with the following open-source InSAR packages: (1) InSAR Scientific Computing Environment 3 (ISCE3) (Rosen et al., 2018) or (2) GMTSAR (Sandwell et al., 2011).

Conflict of Interest

The authors declare there are no conflicts of interest for this manuscript.

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Table 1. Interferograms used for estimating the average postseismic deformation rate over three consecutive periods of interest

S_1	S_2	S_3
April 2007 - October 2007	October 2007 - April 2008	April 2008 - January 2010
04/16/2007-09/01/2007	10/17/2007-01/17/2008	04/18/2008-09/03/2008
04/16/2007-10/17/2007	10/17/2007-03/03/2008	04/18/2008-01/19/2009
09/01/2007-10/17/2007	10/17/2007-04/18/2008	04/18/2008-01/22/2010
	01/17/2008-03/03/2008	09/03/2008-01/19/2009
	01/17/2008-04/18/2008	09/03/2008-01/22/2010
	03/03/2008-04/18/2008	01/19/2009-01/22/2010

Table 2. GPS Validation for InSAR-Derived coseismic Deformation Map

Station	Location	InSAR LOS (cm)	GPS LOS (cm)	Misfit (cm)	Distance from RAVA (km)
RAVA	8.72°S, 157.41°E	7.17	7.17	0	0
MUND	8.33°S, 157.27°E	-86	-89.17	3.17	46.1
HUSU	8.50°S, 157.36°E	-37.54	-33.68	3.86	25.1

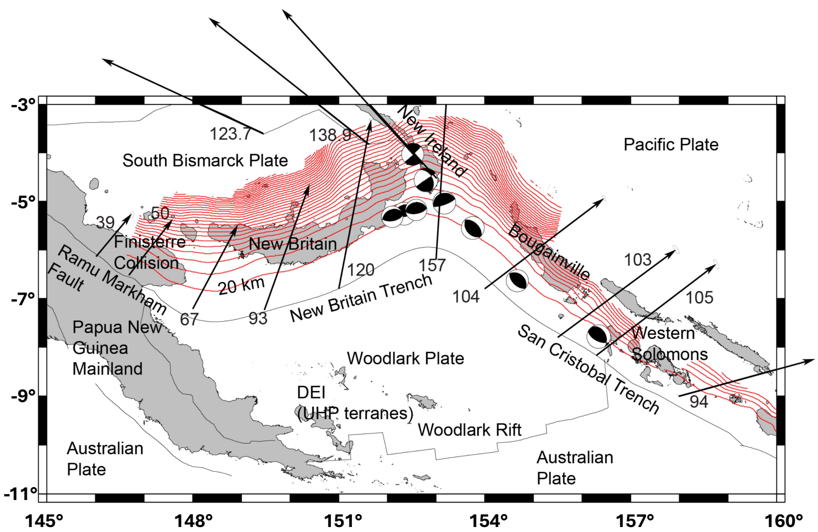

Figure 1. Convergence rates (black arrows, in mm/yr) and slab configuration (red contours) at the Ramu-Markham Fault, New Britain Trench, and San Cristobal Trench. Strike slip rates on the northern boundary of the South Bismarck Plate (e.g. on the Weitin fault that transects New Ireland) are also shown. Beach balls show focal mechanisms for earthquakes larger than Mw 7.5, 1999-present (from Global CMT catalog). New Britain slab contours are in 20 km intervals, with the maximum depth shown at 600 km (contours from Hayes et al. (2012)). West of 153°E, New Britain Trench rates are determined using Woodlark/South Bismarck Plate relative motion (Wallace et al., 2004), while east of 153°E rates are from Woodlark/Pacific Plate relative motion (Wallace et al., 2014). DEI stands for D'Entrecasteaux Island, and UHP stands for Ultra-High Pressure.

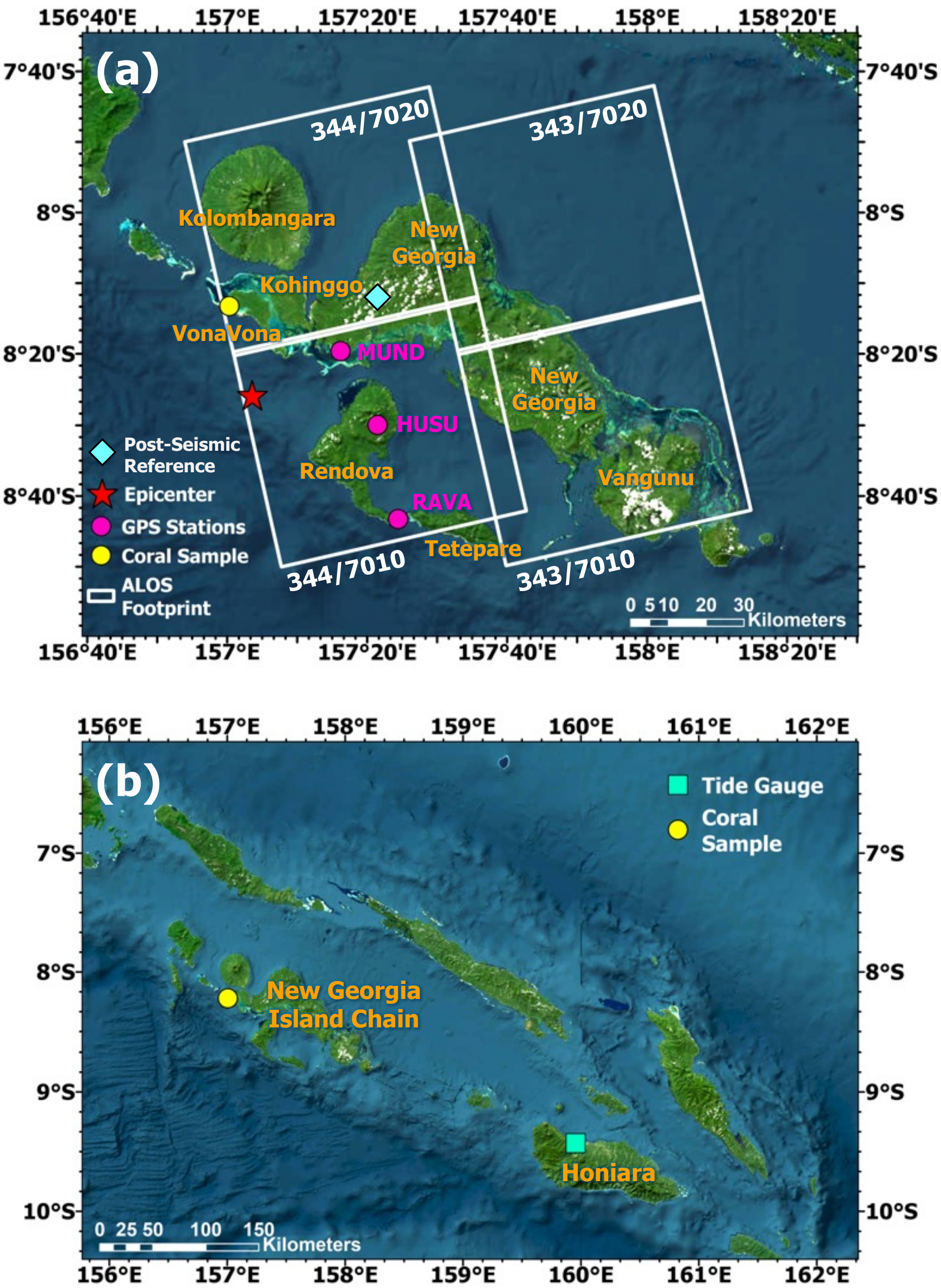

Figure 2. (a) Map of the study area over the Solomon Islands, specifically the New Georgia Island Chain. ALOS PALSAR footprints are outlined in white, and the path/frame numbers are labeled. Locations of three GPS stations, the coral sample site, and the epicenter of the April 1, 2007 Solomon Islands earthquake are labeled according to the legend. (b) The location of the coral sample site near Vonavona in the New Georgia Island chain and the location of the tide gauge off the coast of Honiara.

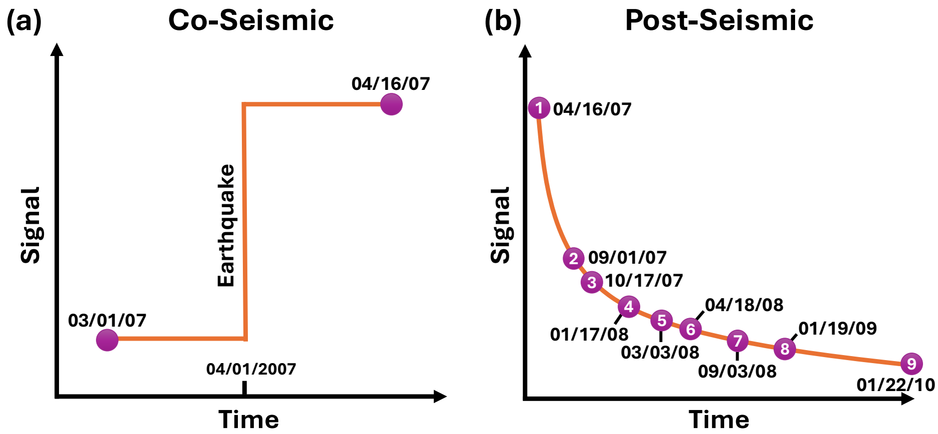

Figure 3. (a) The coseismic signal is modeled as a step function. This transient deformation can be captured by two SAR acquisitions that span the earthquake. (b) The postseismic deformation is expected to follow an exponential or logarithmic decay curve. High-quality Path 344 ALOS PALSAR acquisitions used in the postseismic deformation analysis are marked as purple dots. Note that data gaps are present and the time interval between two nearest acquisitions varies.

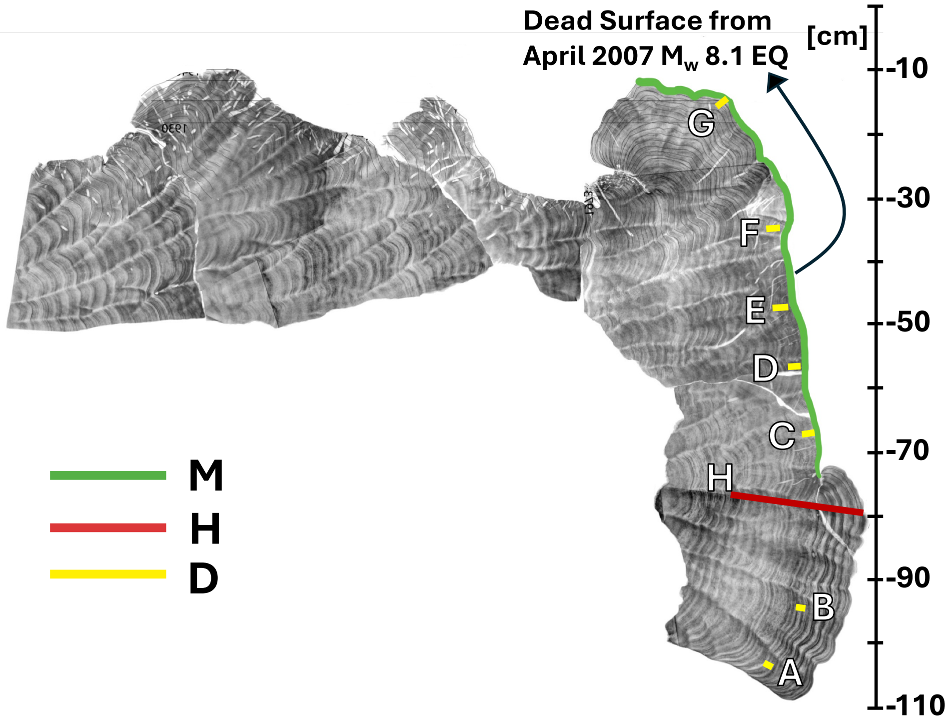

Figure 4. X-ray radiograph of a ∼1 cm-thick slice of fossil *Porites* coral collected in 2012, five years after the April 2007 M_w8.1 earthquake. Growth bands reveal vertical extension rates of 1–1.5 cm/yr. Vertical transects (A–G) were drilled at 1.5 mm spacing to establish the δ^{13}C-depth relationship (marked by the yellow lines with the legend 'D' for depth). Transect H, whose samples were utilized for relative sea level reconstruction between 1996 and 2012, is highlighted by the red line. The notable die-down of the coral surface caused by a large uplift from the 2007 earthquake is highlighted by the green outline, labeled 'M' for morphology.

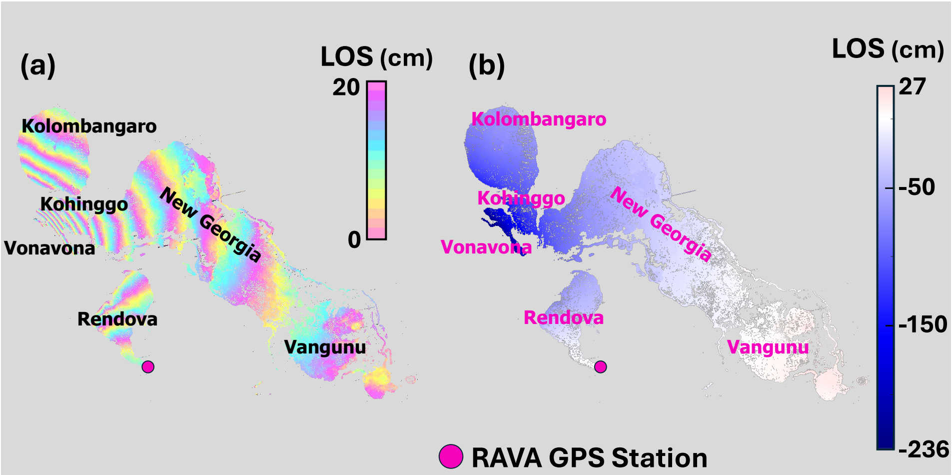

Figure 5. (a) Surface deformation associated with the 2007 Solomon Islands earthquake displayed on a circular color bar. Each fringe of corresponding colors, i.e. pink to pink, represents 20 centimeters of line-of-sight (LOS) deformation. Fringes appear denser near the epicenter. (b) The same InSAR LOS coseismic deformation map on a continuous color map. Blue (negative LOS values) indicates uplift and/or westward motion. The epicenter was identified southwest of Vonavona as shown in Figure 2.

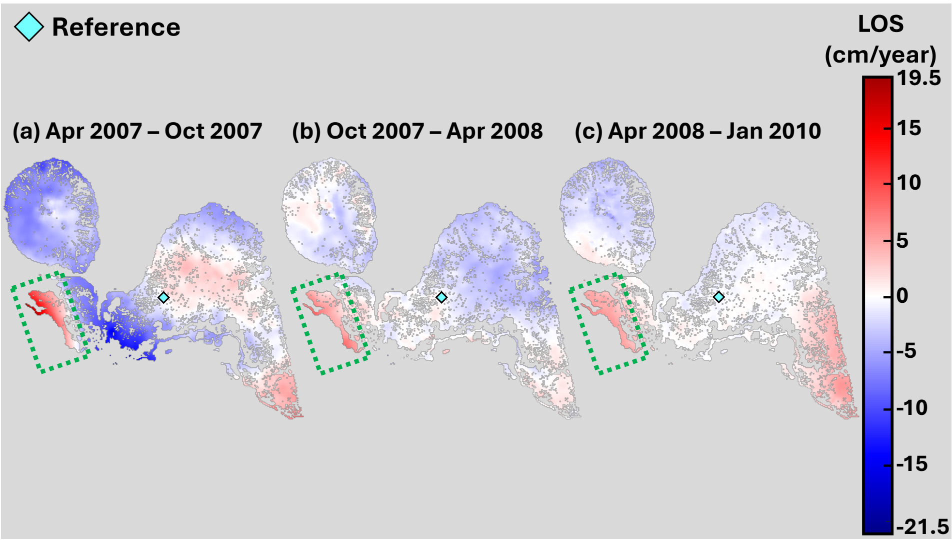

Figure 6. Average postseismic deformation rate estimates (cm/year) for the periods of (a) April 16, 2007 through October 17, 2007; (b) October 17, 2007 through April 18, 2008; and (c) April 18, 2008 through January 22, 2010. Subsidence and/or eastward motion lead to positive LOS motion (red). The LOS deformation rate is relative to the reference pixel located at 8.2°S, 157.4°E marked as a teal diamond. The results over Vonavona, highlighted within the green dashed box, reveal a consistent postseismic deformation pattern with decaying magnitude over time. Outside the green dashed box, the post seismic deformation signals are below the tropospheric noise level and thus undetectable using the ALOS PALSAR interferogram stacks.

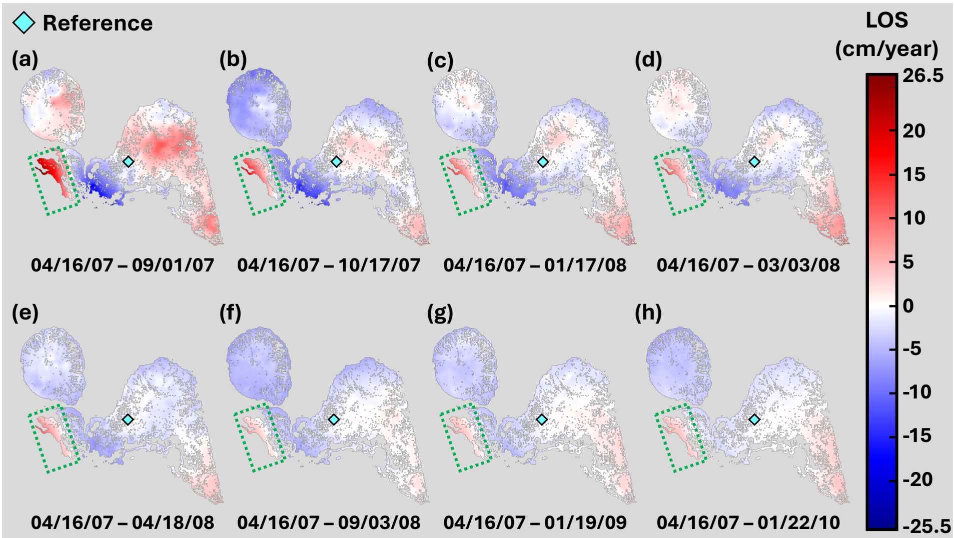

Figure 7. Average postseismic deformation rate estimates (in cm/year) for eight growing time windows starting on April 16, 2007 through: (a) September 1, 2007; (b) October 17, 2007; (c) January 17, 2008; (d) March 3, 2008; (e) April 18, 2008; (f) September 3, 2008; (g) January 19, 2009; and (h) January 22, 2010. Subsidence and/or eastward motion lead to positive LOS motion (red). The reference pixel located at 8.2°S, 157.4°E is marked by a teal diamond. Vonavona, outlined with the green dashed box, maintains a consistent postseismic deformation pattern with a magnitude that decays over time. Outside the green dashed box, the post seismic deformation signals are below the tropospheric noise level and thus undetectable using the ALOS PALSAR interferogram stacks.

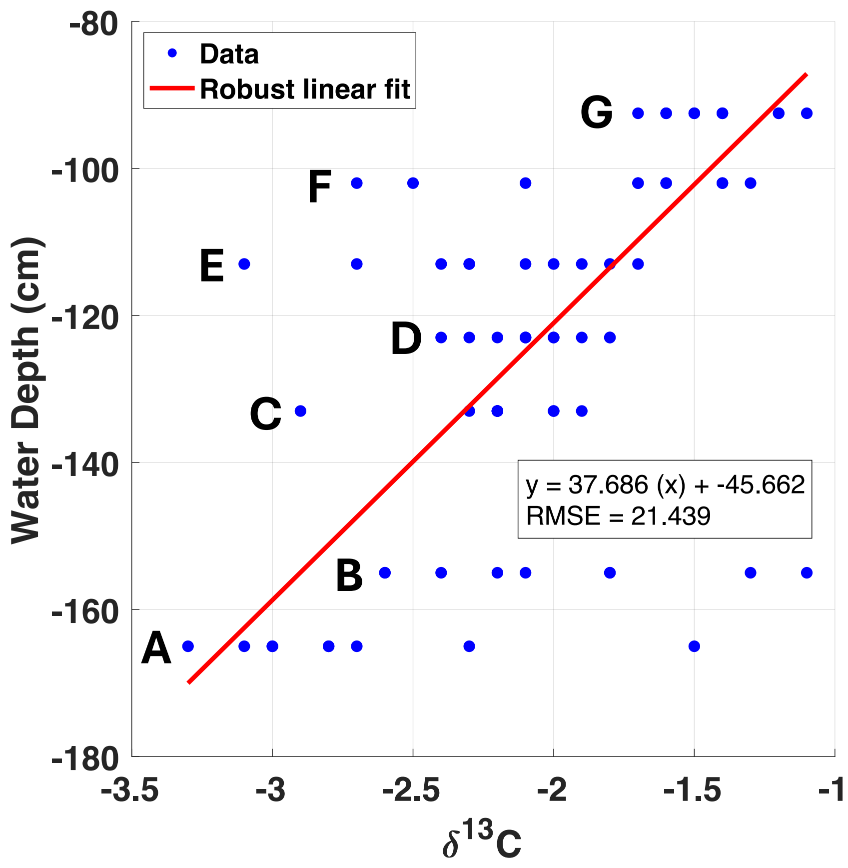

Figure 8. Transfer function between skeletal δ^{13}C and water depth. Blue dots represent δ^{13}C measurements sampled at different water depths. The linear relationship (red line) is used to convert δ^{13}C measurements sampled along transect H in Figure 4 to relative sea level change time series.

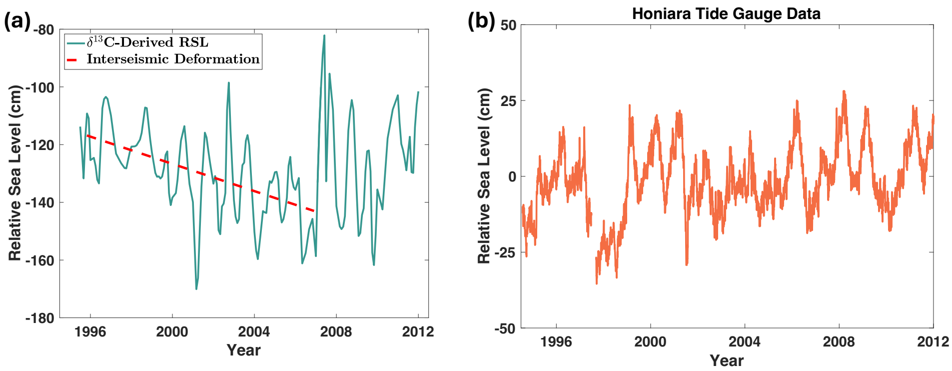

Figure 9. (a) The relative sea level change time series (in cm) as derived from coral δ^{13}C measurements. The red dashed line shows a fitted subsidence trend prior to the 2007 Solomon Island earthquake. A large uplift signal is visible during the earthquake, which is followed by post-seismic subsidence. (b) Changes in relative sea level (cm) over time captured by tide gauge data. The tide gauge data was collected at the Honiara-B station (-9.43° S, 159.96° E), which is far away from the epicenter of the 2007 Solomon Islands earthquake. Both data sets contain similar signals from climate events and seasonal weather variations, while only the coral-derived time series contains vertical tectonic motion associated with the 2007 Solomon Islands earthquake.

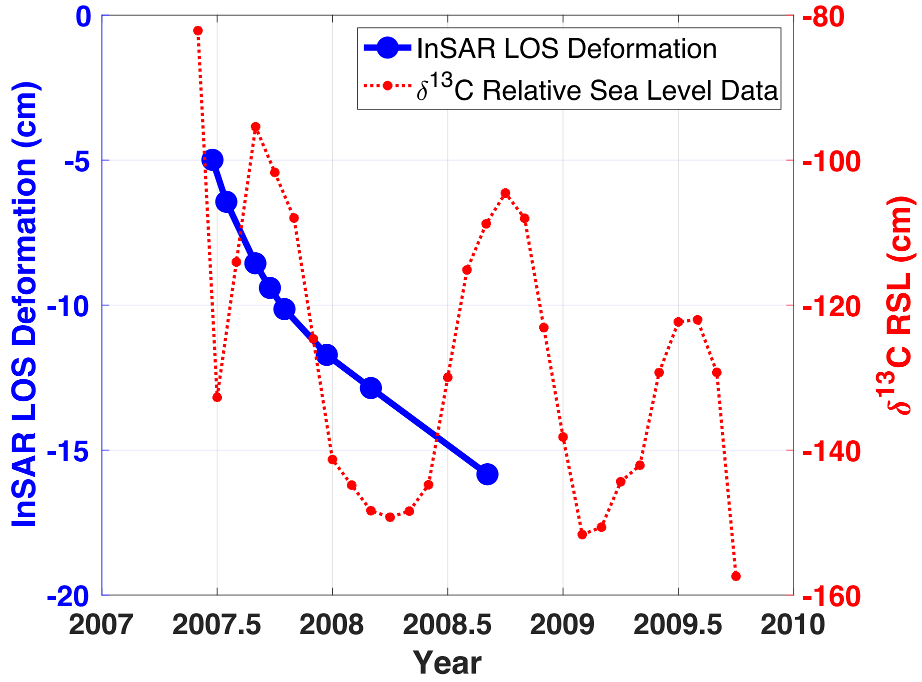

Figure 10. Total postseismic deformation (cm) derived from ALOS PALSAR interferograms (blue dots). The negative sign highlights that the LOS value indicates subsidence of the land. Here the total deformation is derived from the average LOS deformation rates (cm/year) that were estimated over the following time windows spanning April 16, 2007 through: (1) September 1, 2007; (2) October 17, 2007; (3) January 17, 2008; (4) March 3, 2008; (5) April 18, 2008; (6) September 3, 2008; (7) January 19, 2009; and (8) January 22, 2010. Relative sea level time series derived from coral δ^{13}C measurements (red dots) show a similar decay trend.